| Feature | Classical drones | AI-enhanced autonomous drones | Quantum-enabled next-generation drones |
|---|---|---|---|
| **Communication security** | Vulnerable to spoofing and signal interference | Encrypted digital communication with authentication | Quantum key distribution (QKD) and post-quantum cryptography |
| **Fleet/swarm coordination** | Difficult; requires centralized control | Multi-agent coordination with distributed AI | Highly scalable swarm intelligence with real-time collective optimization |
| **Operational environment** | Stable, predictable conditions | Works in partially dynamic environments | Capable of operating in highly dynamic, uncertain, or contested environments |
| **Primary applications** | Photography, hobby use, basic mapping | Agriculture, inspection, logistics, surveillance | Advanced logistics, defense autonomy, disaster intelligence, UAM control networks |

## 9.6 5G and Beyond: Communication Networks for Drones

The evolution of drone capabilities is increasingly shaped by advancements in wireless communication networks, which determine how drones connect, share data, and coordinate operations in real time. Traditional radio and satellite-based control systems limit drone performance due to signal range constraints, limited bandwidth, and high communication latency, especially